Scenario tables for the three uncertain bond choices

Personal research

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The answer

1. **Illiquidity versus mispricing: keep both controls, but they have one blind spot.** In the test grid they cut false trades from 90% to 30%, and catch 59% of true mispricing against 62% for Yu's raw signal. They cannot see illiquidity across a whole issuer, because the sibling bond shares it. Only CDS or trade data can catch that.
2. **Equity delta: real data points to the smaller delta.** On your real CDS panel, the equity-vol delta hedged better overall at 1, 4 and 13 weeks. Your workbook's asset-vol delta *added* variance at 1 week (-14%), against +7% for the equity-vol delta. Neither hedge removes much: the best overall variance cut is 17%.
3. **Fitting $\bar{L}$: the +13% is too kind for low-spread names.** On real names, a +40 bp premium moves the delta by a median +59% under 100 bp, +22% from 100 to 300 bp, and +6% over 300 bp.

Decisions taken for the merged method:

- Keep the peer and sibling checks as coded. Applying the sibling check to adjusted gaps closes one leak, but catches less true mispricing (56% against 59%).
- Replicating your workbook uses the asset-vol delta, because its formulas hold asset vol fixed. The bond method runs with the equity-vol delta as its baseline and reports the asset-vol delta beside it.
- The first bond test excludes names under 100 bp. There both model deltas add variance, and $\bar{L}$ is unstable.

How far to trust this. Tables tagged **BOND** use one made-up issuer and bond, from the worked example. They show how the method responds, not what happens in markets. Tables tagged **REAL CDS** use your master panel: weekly, 2002-09 to 2005-09, 170 names, before costs, in sample, and with GM and Ford already removed. That panel is CDS, not bonds, so it informs the bond choice only indirectly. No bond data has been tested.

To rerun: `python3 scenario_tables.py`, then `python3 build_report.py`. Inputs are `e2c_bond.py`, `worked_example.py`, `e2c_model.py` and `master_panel.csv.gz`. Outputs are one CSV per table in `tables/`, and this document.

Choice 1. Illiquidity versus mispricing

Table 1A, BOND. What each control does. Model Z is 170 bp, and the trigger is $\alpha = 0.5$, so the market must be over 255 bp or under 113 bp. Premia are in bp. "Right answer" uses the true mispricing alone. The CSV holds the same cases for a low-spread and a high-spread issuer.

Case	Right answer	Yu raw	Peer	Sibling	Both	Both right?
Nothing happening	–	–	–	–	–	yes
True mispricing, 100 bp	long	long	long	long	long	yes
True mispricing, 200 bp	long	long	long	long	long	yes
True richness, 60 bp	short	short	short	short	short	yes
Market-wide stress, 100 bp	–	long	–	long	–	yes
Market-wide stress, 200 bp	–	long	–	long	–	yes
This bond illiquid, 100 bp	–	long	long	–	–	yes
This bond illiquid, 200 bp	–	long	long	–	–	yes
Whole issuer illiquid, 100 bp	–	long	long	long	long	**NO**
Whole issuer illiquid, 200 bp	–	long	long	long	long	**NO**
Mispricing 150 in stress 100	long	long	long	long	long	yes
Mispricing 150 in stress 200	long	long	–	long	–	**NO**
Mispricing 150 and bond illiquid 100	long	long	long	long	long	yes

The "NO" rows are the blind spots. Whole-issuer illiquidity passes every control. Large market stress can also hide real mispricing, because subtracting log gaps shrinks it.

Table 1B, BOND. Error rates with measurement noise. Equal weight over 8 mispricing values (three of them zero), 4 market premia and 5 premium types. The rates depend on this grid, which is my choice, so compare controls and do not read the levels as forecasts.

Control	Caught, 0 bp	25 bp	50 bp	False trades, 0 bp	25 bp	50 bp
Yu raw	62%	62%	62%	90%	86%	85%
Peer only	60%	58%	56%	50%	49%	50%
Sibling only	61%	60%	58%	65%	64%	62%
Both, as coded	59%	55%	51%	30%	30%	32%
Both, sibling on adjusted gaps	56%	51%	47%	25%	25%	28%

Table 1C, BOND. What the trade earns under each gap path. Return on capital over the holding period, from the full backtest, with flat stock and rates. Capital is 0.5 per 1 of face. 1x cost is a 1% bond round trip.

What the gap does	150 bp gap, 1x cost	3x cost	300 bp gap, 1x cost	3x cost
closes in 30 days	+11.0%	+6.6%	+24.5%	+19.8%
closes in 60 days	+11.7%	+7.3%	+25.5%	+20.8%
closes in 120 days	+12.8%	+8.4%	+26.8%	+22.1%
never closes	+1.9%	-2.5%	+3.4%	-1.3%
doubles over 180 days	-9.3%	-13.7%	-18.5%	-23.3%

A premium that never closes mostly earns its carry, and higher costs turn that into a loss. The real damage comes if the gap keeps widening.

Table 1D, REAL CDS. Did the peer control help? Overlapping 13-week holds, per 100 of notional, before costs. Hedged with the workbook delta. $\bar{L}$ is fitted on each name's first ten weeks, and the sector median is the peer gap. Equity vol is the workbook blend, not Yu's 1,000-day vol, so this is Yu-style and not a replication. t uses weekly means with Newey-West, 13 lags. A sibling test is not possible here, because each name has one CDS.

Trades	Name-weeks	Share	Mean	Median	Hit rate	t
Yu raw signal, all	13,331	86%	-0.12	-0.06	46%	-0.6
Peer-adjusted signal, all	11,586	74%	+0.05	+0.00	50%	+1.5
Raw signal the peer control removes	3,439	22%	-0.23	-0.13	42%	-1.5
Raw signal, stress weeks (top quartile of median gap)	3,802	24%	+0.06	+0.08	56%	+0.4
Peer-adjusted, stress weeks	3,365	22%	-0.02	-0.00	50%	-0.0
Your workbook signal, all	5,680	36%	+0.10	+0.11	56%	+0.4
Your workbook signal, long credit	3,082	20%	-0.04	+0.05	53%	-0.5
Your workbook signal, short credit	2,598	17%	+0.27	+0.21	60%	+1.1
Raw signal, wide side only (+1)	12,970	83%	-0.13	-0.06	46%	-0.5
Raw signal, tight side only (-1)	361	2%	+0.47	+0.17	54%	+0.9

This is inconclusive. The trades the peer control removes averaged -0.23 (t = -1.5). That points the right way, but no row is statistically significant. The raw signal also fires on 86% of name-weeks, so it barely selects anything. Your workbook's signal is more selective, at 36%. This fixed-hold measure is cruder than your workbook's weekly rebuild, so it does not replace the workbook's own backtest.

Choice 2. Which equity delta

What your files settle. Your workbooks bump the share price with asset vol held fixed (`cds_greeks`, step S12). The recompute matches all 19,489 rows. So your method used the asset-vol delta. That settles what you did, not which is better.

Table 2A, BOND. How far apart the deltas are. Stock hedge per $10mm face. The quoted 2.2x is this grid's base row (2.2x). The gap grows with spread and maturity.

Issuer	S	Equity vol	Years	Model Z (bp)	Asset vol held ($k)	Equity vol held ($k)	Ratio
low spread	60.4	28%	2.0	5	56	41	1.4x
low spread	60.4	28%	4.8	21	344	209	1.6x
low spread	60.4	28%	9.0	45	827	435	1.9x
base	30.2	38%	2.0	126	751	462	1.6x
base	30.2	38%	4.8	170	1,430	638	2.2x
base	30.2	38%	9.0	189	1,836	630	2.9x
high spread	15.1	55%	2.0	619	1,724	849	2.0x
high spread	15.1	55%	4.8	489	1,984	571	3.5x
high spread	15.1	55%	9.0	409	2,013	339	5.9x

Table 2B, BOND. Hedge P&L when the hedge is wrong. Long $10mm of the bond, short the hedge, one instant stock move, in $ thousands. AV is the asset-vol delta and EV is the equity-vol delta. The left three columns assume bond prices really move with asset vol held fixed, and the right three assume equity vol held fixed.

Stock	AV hedge	EV hedge	None	AV hedge	EV hedge	None
-50%	-532	-925	-1,244	+177	-216	-535
-30%	-163	-399	-590	+177	-58	-249
-20%	-67	-224	-352	+134	-23	-151
-10%	-16	-94	-158	+73	-5	-69
+10%	-14	+65	+129	-83	-4	+59
+20%	-51	+106	+234	-173	-16	+111
+30%	-107	+129	+320	-270	-34	+157
Default	-3,638	-4,423	-5,061	-3,638	-4,423	-5,061

Each hedge works in its own world, and fails in the other. In default, either hedge leaves most of the loss in place: the bond loses about $5,061k, and the better hedge recovers only about $1,423k.

Table 2C, REAL CDS. Which delta hedged better. Long protection plus long stock, with no rebalancing. AV and EV are the asset-vol and equity-vol deltas. "Var cut" is the fall in variance from adding the hedge, and "Abs cut" is the fall in mean absolute P&L. A negative number means the hedge made things worse. "Fitted slope" is an in-sample flat hedge, for reference.

Weeks	Spread	N	AV delta	EV delta	Fitted slope	Var cut AV	Var cut EV	Abs cut AV	Abs cut EV
1	All	16,960	0.060	0.037	0.035	-14%	+7%	-33%	-7%
1	under 100 bp	13,362	0.050	0.032	0.008	-72%	-20%	-59%	-25%
1	100 to 300 bp	2,773	0.131	0.067	0.054	-11%	+6%	-20%	+1%
1	over 300 bp	825	0.225	0.081	0.117	-8%	+12%	-16%	+5%
4	All	12,235	0.059	0.036	0.051	+12%	+16%	-10%	+4%
4	under 100 bp	9,782	0.050	0.032	0.012	-58%	-12%	-36%	-9%
4	100 to 300 bp	1,924	0.130	0.067	0.081	+7%	+17%	+3%	+12%
4	over 300 bp	529	0.221	0.085	0.183	+22%	+17%	+12%	+11%
13	All	4,837	0.059	0.037	0.045	+10%	+17%	+1%	+11%
13	under 100 bp	3,933	0.051	0.033	0.019	-47%	-4%	-15%	+5%
13	100 to 300 bp	703	0.126	0.067	0.076	-4%	+15%	+10%	+14%
13	over 300 bp	201	0.210	0.088	0.177	+11%	+16%	+20%	+18%

Under 100 bp, which is 79% of the 1-week sample, both model deltas are far above the fitted slope and mostly add variance. Over 300 bp the result is mixed, and the asset-vol delta sometimes does better.

Choice 3. Fitting $\bar{L}$ to bond spreads

Table 3A, BOND. A premium in the fit window. The fit absorbs a liquidity premium into $\bar{L}$, which moves the hedge. The +40 and -40 rows reproduce the quoted +13% and -16%. Hedge figures are per $10mm face. "Fit status" is the code's own flag: above $\bar{L} = 1$ it asks for review, and at the bound it rejects.

Premium (bp)	$\bar{L}$	Fit status	Model Z	Delta change	Stock hedge ($k)	Hedge error ($k)	5y default prob.
-120	0.29	ok	50	-60%	575	-848	5%
-80	0.45	ok	90	-36%	909	-513	9%
-40	0.63	ok	130	-16%	1,189	-234	13%
-20	0.73	ok	150	-8%	1,311	-111	14%
+0	0.84	ok	170	+0%	1,423	+0	16%
+20	0.96	ok	190	+7%	1,523	+101	18%
+40	1.08	review	210	+13%	1,613	+190	19%
+80	1.33	review	250	+24%	1,761	+339	22%
+120	1.61	review	290	+31%	1,870	+447	25%
+200	2.21	review	371	+39%	1,984	+561	29%
+300	3.00	reject	466	+40%	1,986	+563	35%

Table 3B, BOND. False signals when the premium changes after the fit. There is no mispricing anywhere in this table, so **every signal is false**. Cells show market minus model in bp. L or S marks a long-bond or short-bond signal at $\alpha = 0.5$. Lower-case l or s marks a signal that fires only at $\alpha = 0.25$.

Premium in fit	-40 later	+0 later	+40 later	+80 later	+120 later	+200 later	+300 later
-40	+0	+40 l	+80 L	+120 L	+160 L	+240 L	+340 L
+0	-40 s	+0	+40	+80 l	+120 L	+200 L	+300 L
+40	-80 S	-40	-0	+40	+80 l	+160 L	+260 L
+80	-120 S	-80 s	-40	-0	+40	+120 l	+220 L
+120	-160 S	-120 S	-80 s	-40	-0	+80 l	+180 L
+200	-241 S	-201 S	-161 S	-121 s	-81 s	-1	+99 l

So a change of about 120 bp in the premium is enough to fire a false signal at $\alpha = 0.5$ on this bond, and about 80 bp at $\alpha = 0.25$.

Table 3C, REAL CDS. The same premium test on real names. A premium is added to each name's first ten weekly spreads, then $\bar{L}$ is refitted and the delta recomputed. "At bound" is the share of fits that hit the code's limits. These ten weeks fall in late 2002, a stressed period, which may affect the fits.

Spread	Premium (bp)	Names	Median delta change	10th pct	90th pct	$\bar{L}$ before	$\bar{L}$ after	At bound
under 100 bp	-40	67	-66%	-91%	-47%	0.73	0.33	1%
under 100 bp	-20	92	-39%	-77%	-23%	1.03	0.68	2%
under 100 bp	+20	93	+32%	+17%	+87%	1.05	1.33	14%
under 100 bp	+40	93	+59%	+27%	+192%	1.05	1.57	22%
under 100 bp	+80	93	+101%	+27%	+301%	1.05	2.05	35%
100 to 300 bp	-40	51	-24%	-35%	-11%	0.42	0.31	0%
100 to 300 bp	-20	51	-12%	-17%	-5%	0.42	0.36	0%
100 to 300 bp	+20	51	+11%	+4%	+18%	0.42	0.49	0%
100 to 300 bp	+40	51	+22%	+8%	+35%	0.42	0.56	4%
100 to 300 bp	+80	51	+41%	+15%	+67%	0.42	0.73	4%
over 300 bp	-40	26	-7%	-10%	+0%	0.56	0.44	8%
over 300 bp	-20	26	-3%	-5%	+0%	0.56	0.50	8%
over 300 bp	+20	26	+3%	-0%	+5%	0.56	0.64	8%
over 300 bp	+40	26	+6%	-0%	+9%	0.56	0.71	8%
over 300 bp	+80	26	+12%	-0%	+18%	0.56	0.84	8%